

• Time : 2 hr 15 minutes

• Max Marks : 33

1. An electromagnetic wave in vacuum is given (in SI units) by

$$\mathbf{E}(\mathbf{r}, t) = (-6\hat{i} + 3\sqrt{5}\hat{j}) \times 10^4 \text{ V/m} \exp \left\{ i \left[\frac{\pi}{3} (\sqrt{5}x + 2y) \times 10^7 - 9.42 \times 10^{15}t \right] \right\}.$$

Assuming this represents a physical plane wave:

- (a) Find the direction of oscillation of the electric field (unit vector). [1+1+2+1+2]
 (b) Find the scalar amplitude E_0 of the electric field.
 (c) Determine the direction of propagation (unit vector along $\mathbf{k}$).
 (d) Obtain the magnitude of the wave vector k and the wavelength λ .
 (e) Obtain the speed v of the wave.
2. An electromagnetic plane wave of angular frequency ω propagates in a homogeneous conducting medium characterized by ϵ , μ , and conductivity σ . Assume all fields vary as $e^{i(kz - \omega t)}$. [4+3+1]

- (a) Starting from Maxwell's equations, derive the dispersion relation

$$k^2 = \mu\epsilon\omega^2 \left(1 + i \frac{\sigma}{\epsilon\omega} \right).$$

- (b) In the good conductor limit $\sigma \gg \epsilon\omega$, show that the complex wave number can be written as

$$k \approx (1 + i) \sqrt{\frac{\mu\sigma\omega}{2}}.$$

(Hint: You can use $i = e^{i\pi/2}$). Hence, obtain the skin depth

$$\delta = \sqrt{\frac{2}{\mu\sigma\omega}}$$

and explain briefly its physical meaning.

- (c) Comment qualitatively on what happens to δ as the frequency ω increases.
3. For light incident from a lossless dielectric of refractive index n_i onto another lossless dielectric of index n_t , the Fresnel amplitude coefficients for perpendicular (s-) polarization at an angle of incidence θ_i are

$$r_{\perp} = \frac{E_r}{E_i} = \frac{n_i \cos \theta_i - n_t \cos \theta_t}{n_i \cos \theta_i + n_t \cos \theta_t}, \quad t_{\perp} = \frac{E_t}{E_i} = \frac{2n_i \cos \theta_i}{n_i \cos \theta_i + n_t \cos \theta_t},$$

where θ_i and θ_t are related by Snell's law.

- (a) Starting from the definitions

$$R_{\perp} = \frac{I_r}{I_i}, \quad T_{\perp} = \frac{I_t}{I_i},$$

and using the correct expressions for the normal component of the time-averaged Poynting vector in each medium, $I \propto n \cos \theta |E|^2$, explicitly derive formulas for $R_{\perp}$ and $T_{\perp}$ in terms of $r_{\perp}$ and $t_{\perp}$, and then show by direct algebra that for non-absorbing media

$$R_{\perp} + T_{\perp} = 1.$$

(b) Starting from the Fresnel expressions above, show that the identity

$$t_{\perp} + (-r_{\perp}) = 1$$

holds for all angles of incidence, and briefly discuss its physical interpretation in terms of field continuity at the interface.

4. A taut string of linear mass density μ extends along the x -axis. For $x < 0$ the string obeys the ordinary wave equation, while for $x > 0$ the string is coated with a damping material and obeys

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2} - \gamma \frac{\partial y}{\partial t},$$

where γ is a constant damping coefficient. A harmonic wave

$$y_i(x, t) = A e^{i(kx - \omega t)}, \quad k = \frac{\omega}{v},$$

is incident on the interface at $x = 0$ from the left.

[4+4+2]

- (a) Apply the boundary conditions of continuity of displacement and transverse force at $x = 0$ to derive the reflection and transmission coefficients for the wave at the interface. (Hint: For the wave in the damped region, you can use the solution: $C e^{i(qx - \omega t)}$, where q is, in general, complex.)
- (b) In the limit of weak damping ($\gamma \ll \omega$), show that the transmitted wave has (i) an exponentially decaying amplitude and (ii) an approximately unchanged wavelength. Explicitly find the attenuation coefficient.
- (c) Determine the fraction of incident power reflected back into the undamped region.